Lesson plan · for the teacher

The shortest way

Optimization · Level 1 · Lesson 2

Time
40 min

Materials
Chalk (or string and paper labels indoors); Number cards for each village; The map from the lesson, copied on the board

Learning goals

- Find a way's length by adding up its roads.
- Find the shortest way by settling the nearest place first.
- Find a new way when a road closes, and ask who walks the farthest.

Activity: Playground Dijkstra

1. Warm-up (5 min): Draw the lesson's map on the board. Ask pupils to guess the shortest way by looking, and write the guesses down.
2. Teach (8 min): Add up two ways together (15 km through Bora, 13 km through Tala, Kesi and Mina). Explain the rule: settle the waiting place with the smallest number, then lower its neighbours' numbers.
3. Chalk map (15 min): Chalk the map on the playground. One pupil stands on each village with a number card. Home says 0. Each turn, the class names the waiting village with the smallest number; its pupil sits down (settled), and neighbours change their cards if the new way is shorter.
4. Flood a road (7 min): Rub out the bridge between Kesi and Mina. Start again from home: the new shortest way is through Rubi and Sana, 14 km.
5. Wrap-up (5 min): Compare with the guesses. Ask which pupils' real way to school is longest, and what would help them.

No playground? Lay string between chairs as roads and tape paper labels with the lengths. Keep the same map as the lesson so pupils can check their answers: 13 km, then 14 km.

Discussion

- Why did the straight-looking road lose?
- Why can a settled village never get a smaller number?
- Who walks the farthest to our school, and what would help?

Exit ticket

- Draw a small map with four places and road lengths, and mark the shortest way.

Try it in class tomorrow
Chalk a map on the playground: home, a few villages and the school, with a length on every road. Pupils stand on the villages with number cards and settle the smallest number each turn until the school is settled. Then close a road and find the new shortest way together.

Worksheet · for pupils

The shortest way

Optimization · Level 1 · Lesson 2

Name _____ Class _____ Date _____

1 **Mr Okafor's pupils say the road through Bora must be the shortest way to school, because it looks the straightest on the map. Are they right?**

- ◯ Only when the bridge is flooded
- ◯ No: through Tala, Kesi and Mina adds up to less
- ◯ Yes: straight roads are always shortest

2 **Put Dijkstra's method in order.**

- ☐ Settle the waiting place with the smallest number
- ☐ Give home 0 and the other places no number yet
- ☐ Follow the roads back to read the route
- ☐ Use its roads to lower its neighbours' numbers
- ☐ Repeat until the school is settled

3 **In Dijkstra's method, once a place is settled, a shorter way to it can still turn up later.**

- ◯ True
- ◯ False

4 **The bridge between Kesi and Mina is flooded. What is the shortest way from home to school now?**

- ◯ Home, Tala, Kesi, School: 15 km
- ◯ Home, Rubi, Sana, School: 14 km
- ◯ Home, Bora, School: 15 km

5 **A road's "length" can be anything you add up. Match each job to the number to put on its roads.**

The quickest drive to the clinic	• • • •	Fares in coins
The cheapest bus trip to the city	• • • •	Minutes
The safest walk for small children	• • • •	Kilometres
The shortest walk to the well	• • • •	Busy crossings

Answers · for the teacher

The shortest way

Optimization · Level 1 · Lesson 2

1 **No: through Tala, Kesi and Mina adds up to less**

Add the roads, don't trust the look: through Bora is 5 + 10 = 15 km, while the winding way through Tala, Kesi and Mina is 4 + 3 + 2 + 4 = 13 km.

2 **Top to bottom: 2, 1, 5, 3, 4**

Start from home at 0, then go round: settle the smallest, update its neighbours.

3 **False**

A place is settled only when it has the smallest number of all the waiting places.

4 **Home, Rubi, Sana, School: 14 km**

Without the bridge, the old 13 km way is gone.

5 **The quickest drive to the clinic — Minutes**
The cheapest bus trip to the city — Fares in coins
The safest walk for small children — Busy crossings
The shortest walk to the well — Kilometres

The method works the same whatever you add up, as long as no number is negative.

For you to think about

Which journey in your pupils' lives could be planned better with a map and road lengths: the way to school, to the well, to the clinic? Who walks the farthest?